

Capacitively-Coupled Chopper Instrumentation
Amplifier for Linear Hall Effect Sensor
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Team Chop

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Project Information

Project Name	CCIA for Linear Hall Effect Sensor
Team Members	alang
Review Date	
Review Version	
Track	B

Design Objective

The aim of this project is to design and characterize a monolithic magnetic sensing system based on the Hall Effect. If tapeout is completed, the specifications of the sensor and readout circuit will be characterized individually.

The target specifications for the magnetic sensing system are shown in Table 1. See the Appendix for the full working specification documents for all blocks.

Specification	Value
Maximum Offset	100 μT
Area	0.25 mm^2
Resolution	100 μT_{rms}
Input Range	$\pm 100 \text{ mT}$
Bandwidth	10 kHz
Supply Voltage	3.3 V
Supply Current	5 mA
Temperature Range	-40°C to 125°C
Clock Frequency	100 kHz
Output Common-Mode Voltage	1 V

Table 1: System Specifications

The motivation for this project in the context of the Chipathon program is to provide reusable IP to the open-source silicon community. Priority will be given to designing a simple system that can be well characterized as opposed to a novel system. Overall, if the tapeout is successful, future designers would be able to reuse and customize both the sensor design and the readout circuit for their specific application. In any case, future Chipathon participants will be able to learn from the successes and failures of the project.

System Overview

A block diagram of the system is shown in Fig. 1.

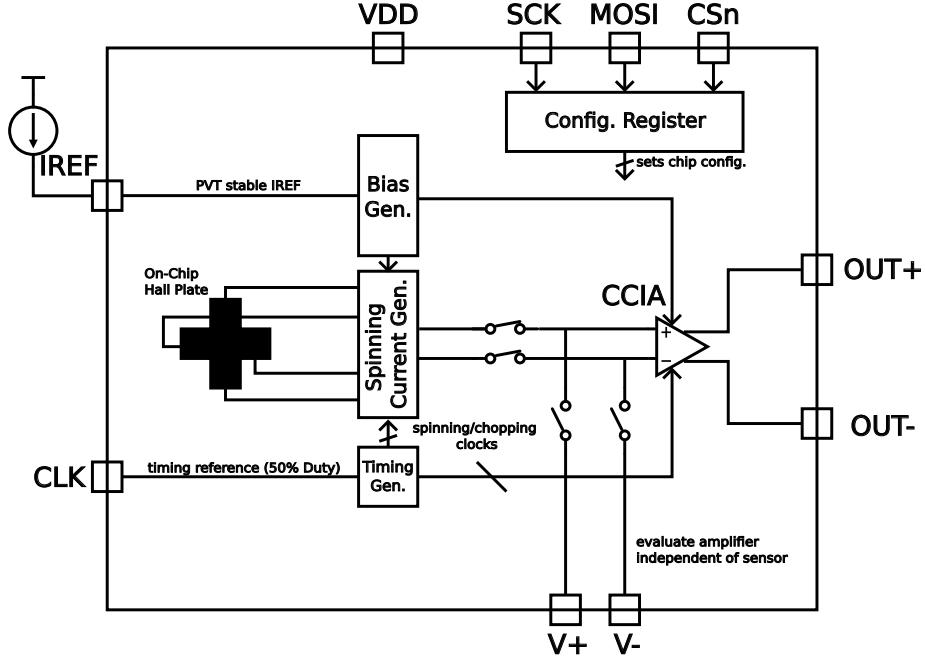


Figure 1: Block diagram of the proposed architecture.

In the diagram, an on-chip horizontal Hall plate converts an incoming magnetic field (perpendicular to plane of chip) into a small voltage that is amplified by the instrumentation amplifier (IA) prior to being driven to the DUT output pins (OUT+/OUT-). The entire design operates from a single supply. Aside from the SPI configuration interface (only used during idle periods), the entire design operates in a single synchronous clock domain derived from the external clock signal (CLK). Two additional pins are used as analog debug pins (V+/V-). Based on the system configuration, these pins will be used to probe various points within the design to enable individual characterization of the sensor and IA.

A full pinout table of the design is given in Table 2.

Analog Pins		
IREF	Analog	Current reference
OUT+	Analog	(+) Output
OUT-	Analog	(-) Output
VTP+	Analog	(+) Test Point
VTP-	Analog	(-) Test Point
Digital Pins		
SCK	Digital Input	SPI configuration clock
MOSI	Digital Input	SPI configuration data input
CSn	Digital Input	SPI chip select
CLK	Digital Input	Chopping/spinning clock reference
Supply Pins		
VDD	Power	Supply input
VSS	Power	Ground (shared with other Chipathon blocks)

Table 2: Project I/O description.

Schematic Summary

This section describes the schematic-level operation of each part of the design. For each block, both behavioral and transistor-level models are described, including motivation for key design decisions. For all blocks, the transistor-level sizing is subject to change during the verification process.

On-Chip Hall Plate

- Purpose
- Inputs and Outputs
- Sub-sheets
- Behavioral Model
- Design Methodology
- Transistor-Level Implementation

Bias Generation

- Purpose
- Inputs and Outputs
- Sub-sheets
- Behavioral Model
- Design Methodology
- Transistor-Level Implementation

Spinning Current Generation

- Purpose
- Inputs and Outputs
- Sub-sheets
- Behavioral Model
- Design Methodology
- Transistor-Level Implementation

Timing Signal Generation

- Purpose
- Inputs and Outputs
- Sub-sheets
- Behavioral Model
- Design Methodology
- Transistor-Level Implementation

Configuration Register

- Purpose
- Inputs and Outputs
- Sub-sheets
- Behavioral Model
- Design Methodology
- Transistor-Level Implementation

Instrumentation Amplifier

- Purpose
- Inputs and Outputs
- Sub-sheets
- Behavioral Model
- Design Methodology
- Transistor-Level Implementation

Design Assumptions

The following assumptions are made within the design.

- The circuit shall be designed to operate from a single 3.3V supply
- The circuit shall be designed to operate over the temperature range of (-40°C - 125°C)
- The specifications of each circuit shall be simulated over all worst-case process corners
- Each analog output (both (+) and (-)) of the readout circuit shall be capable of driving a 10 pF off-chip load
- To test the full system, an external magnetic field will be applied via a Helmholtz coil pair. A printed circuit board will be designed to supply all DUT inputs and condition the differential readout circuit output prior to digitization.

Verification Status

Functional Simulations

Corner Simulations

Monte Carlo

Sign-off

- ERC Status
- DRC Status
- LVS Status

Design Checklist

Functionality

- Requirements implemented
- Reset behavior verified
- Startup verified
- Power sequencing considered

Analog

- Bias currents checked
- Device operating regions verified
- Stability analyzed
- Noise considered
- Matching requirements identified

Digital

N/A

Mixed-Signal

N/A

Reliability

- Device voltage limits checked
- Current density reviewed
- Latch-up prevention considered
- ESD strategy identified

Documentation

- Nets clearly named
- Hierarchy consistent
- Parameters documented
- Symbols match implementation

Open Issues

List of open issues with:

- Description
- Impact
- Proposed Solution
- Owner

Questions for Reviewers

Review Outcome

Appendix

[1]

References

- [1] L. Lamport, *LaTeX: A Document Preparation System*, 2nd ed. Addison-Wesley, 1994.